

AEROBIC AND ANAEROBIC CHANGES WITH HIGH-INTENSITY INTERVAL TRAINING IN ACTIVE COLLEGE-AGED MEN

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ABSTRACT

Ziemann, E, Grzywacz, T, Łuszczuk, M, Laskowski, R, Olek, RA, and Gibson, AL. Aerobic and anaerobic changes with high-intensity interval training in active college-aged men. *J Strength Cond Res* 25(4): 1104–1112, 2011—We investigated the aerobic and anaerobic benefits of high-intensity interval training performed at a work-to-rest ratio of 1:2 because little performance enhancement data exist based on this ratio. Recreationally active male volunteers (21 years, 184 cm, 81.5 kg) were randomly assigned to a training (interval training [IT] $n = 10$) or control group ($n = 11$). Baseline assessments were repeated after the last training session. Each participant underwent basic anthropometric assessment and performed a $\dot{V}O_2$ max test on an electronically braked cycle ergometer and a 30-second Wingate test. Venous samples were acquired at the antecubital vein and subsequently processed for lactate (LA); samples were obtained at rest, and 5 and 15-minute post-Wingate test. The interval training used a cycling power output equivalent to 80% of $\dot{V}O_2$ max (80% $\rho \dot{V}O_2$ max) applied for 6 90-second bouts (each followed by 180-second rest) per session, 3 sessions per week, for 6 weeks. The control group maintained their normal routine for the 6-week period. Group $\times$ time repeated-measures analyses of variance revealed that IT improved $\dot{V}O_2$ max (5.5 ml·kg⁻¹·min⁻¹), anaerobic threshold (3.8 ml·kg⁻¹·min⁻¹), work output (12.5 J·kg⁻¹), glycolytic work (11.5 J·kg⁻¹), mean power (0.3 W·kg⁻¹), peak power (0.4 W·kg⁻¹), and max power (0.4 W·kg⁻¹); $p < 0.05$. Posttesting LA was lower on average for IT at the 5-minute mark but significantly so at the 15-minute mark. Twenty-seven minutes of cycling at 80% $\rho \dot{V}O_2$ max applied with a work-to-rest ratio of 1:2 and spread over 3 sessions per week for 6 weeks provided sufficient stimulus to significantly improve markers of anaerobic and

aerobic performance in recreationally active college-aged men. Inclusion of such a protocol into a training program may rapidly restore or improve a client's or athlete's maximal functional capacity.

KEY WORDS blood lactate, Wingate, Training adaptations, cycling

INTRODUCTION

Short duration, high-intensity interval training (HIIT) has been proposed as an effective way to improve aerobic capacity (11–13,27) and found to be beneficial when training for endurance sports such as running, cycling, and swimming (12,21,26–28). Diverse HIIT protocols are well documented (13,20,26), and extensive reviews of training adaptations to short-sprint (33) and interval training (6,7) detail the variety in volume, duration, recovery, frequency, or sample (33).

According to Daniels and Scardina (17), short-duration work intervals performed at or near velocities associated with $\dot{V}O_2$ max can prolong the time at which intensities of 95–100% $\dot{V}O_2$ max can be maintained during a training session (17). Consequently, previous studies have mostly focused on work-to-rest ratios of 1:1 or greater (2:1, 4:1) with interval durations ranging from 15 seconds to 2.5 minutes (8,34). In these studies, an increase in $\dot{V}O_2$ max and the velocity associated with $\dot{V}O_2$ max has been presented. Additionally, various passive and active recovery periods after exertion at intensities of 50–70% $\dot{V}O_2$ max combining to make work-to-rest ratios of 1:1, 2:1, and 4:1 have been examined. The results obtained in these studies confirmed a beneficial impact of interval training while at the same time presenting limitations to completion of the entire training program when the exercise intensity was too high and the recovery time too short even though the subjects had an excellent level of physical conditioning (8,34).

In various sport disciplines, training with a work-to-rest ratio of 1:2 or 1:3 is often recommended to prepare for competition that requires a similar physical effort and development of particular energetic systems. For example, randori training (fighting practice) in judo uses this pattern of

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work-to-rest ratio (14). A work-to-rest ratio of 1:2 is also specific for ice-hockey players during the game (25). Time-motion analysis of an ice-hockey match indicated that the average shift lasts about 85 seconds with 225-second recoveries between shifts (22). In tennis, recent research has shown that the duration of work and rest periods during a match is 5–10 and 10–20 seconds, respectively, for a work-to-rest ratio in the range of 1:1 and 1:2 to 1:4 (18).

Sport-specific training programs are designed to optimize players' skills and should also mimic the physiological and energetic demands of a particular sport. For most sports, the adenosine triphosphate-phosphocreatine (ATP-PC), glycolytic, and oxidative phosphorylation systems play important roles. The first 2 energy systems are the main sources of ATP during high-intensity exercise, whereas the mitochondrial system plays a crucial role in terms of recovery periods.

Most of the published reports focus mainly on the changes in aerobic capacity as a result of HIIT with a work-to-rest ratio of 1:2. Fox suggested that a work-to-rest ratio of 1:2 is optimal for improving the glycolytic system (19). Recently, Laursen et al. (27) confirmed this observation, reporting that 4 weeks of HIIT increased anaerobic capacity as evaluated through accumulated oxygen deficit. The main goal of the present study was to determine if a 6-week training program using a work-to-rest ratio of 1:2 with passive rest periods between bouts could improve both anaerobic and aerobic capacities in a sample of recreationally active, college-aged men. We hypothesized that a protocol using 6 90-second cycling bouts at 80% $\dot{V}O_{2\max}$ (power at 80% $\dot{V}O_{2\max}$) separated by 180-second passive recovery periods over a 6-week period would significantly enhance both anaerobic (lactic and alactic) and aerobic capacity.

METHODS

Experimental Approach to the Problem

An overview of the experimental protocol is presented in Table 1. Participants (Table 2) completed the same battery of body composition, aerobic, and anaerobic assessments before and after the intervention period. The exercise tests were

performed on separate days before and after the HIIT intervention. Aerobic measurements were performed 5 days before and 2 days after the training intervention; anaerobic measurements were repeated 2 days after the aerobic assessments. The variables of interest included those indicative of maximal aerobic performance (Table 3) and maximal anaerobic performance (Table 4) as determined through graded exercise cycling trials to volitional exhaustion and a maximal exertion 30-second Wingate anaerobic test (WAnT), respectively. Blood lactate (LA) responses (Table 5) were investigated during rest and at 2 time intervals after the final WAnT. These variables were selected because of their universal applicability to the research question, ease of study replicability, financial constraints, and the logistical need to reduce the level of invasiveness. Based on earlier reports, the combined measurements of the mechanical power and underlying anaerobic and aerobic energy metabolism have clearly demonstrated that, for adults, the metabolic profile of the WAnT is highly anaerobic (ca. 80%) with a minor (ca. 20%) aerobic component (1,3–5,15,24,30,32,36,37). Using the Multi Cyclo Ergometer System (MCE v 5.1 Software, Sport Institute, Warsaw, Poland), the phosphagenic and nonphosphagenic components of work were determined from the 30-second Wingate sprint data for each participant. The basic anaerobic parameters achieved with MCE and the original Wingate software have shown a strong correlation ($r = 0.98$). The identification of the phosphagenic and nonphosphagenic components is an authorized complement of the MCE 5.1 version, previously investigated by our team in collaboration with Sport Institute (38).

Subjects

Twenty-one healthy, physically active but not highly trained, college-aged men volunteered to take part in the experiment. The participants were fully informed of the risks and stresses associated with the study and gave written informed consent to participate in this study as approved by the Medical Research Ethics Committee of the Academy of Physical Education and Sport. Each participant underwent

TABLE 1. High-intensity interval training investigational timeline.*

I. Protocol before interval training				
A	Anthropometric measurements, $\dot{V}O_{2\max}$ test aerobic performance	2-d rest	WAnT anaerobic performance	3-d rest
B	(6 wk/3 times/wk); 6 $\times$ 90 s 80% $\dot{V}O_{2\max}$ work with 180-s rest†			
II. Protocol after interval training				
C	2-d rest	Anthropometric measurements, aerobic performance	3-d rest	Anaerobic performance

*HIIT = high-intensity interval training

†Total single session of exercise values was 155 kJ; 1-week session was 465 kJ and total training program was 2,795 kJ.

TABLE 2. Participant demographic and anthropometric characteristics.*†‡

Variable	Pretraining		Posttraining	
	IT group	C group	IT group	C group
Age (y)	21.6 ± 1.1	21.0 ± 0.9		
Height (cm)	182.5 ± 2.5	186.0 ± 5.7		
Weight (kg)	81.6 ± 6.2	81.4 ± 7.1	82.0 ± 5.3	81.7 ± 7.0
FFM (kg)	68.2 ± 4.6	69.9 ± 5.7	68.6 ± 4.1	70.0 ± 5.4
Fat (kg)	13.4 ± 3.2	11.0 ± 3.0	13.5 ± 2.2	11.6 ± 2.8
Fat %	16.2 ± 3.2	13.5 ± 3.0	16.4 ± 2.2	14.1 ± 2.7
BMI (kg·m ⁻²)	24.5 ± 1.8	23.0 ± 1.9	24.6 ± 1.5	23.6 ± 2.0

*Fat = fat mass; FFM = free fat mass; BMI = body mass index; IT = interval training group; C = control group.

†Values are means ± SD.

‡Values are not significantly different for group or time.

preliminary exercise testing to become familiar with the exercise model and to obtain a measure of their maximal aerobic work capacity 1 week before the experimental trial. The participants were randomly assigned to either the interval training (IT; $n = 10$) or control (C; $n = 11$) group.

The first stage of the study (baseline measurements and first 3 weeks of training) was conducted during the semester break. The last 3 weeks of training was performed during the first 3 weeks of the semester. Nevertheless, the 6 weeks of the protocol were contiguous. The participants' class schedules included both theoretical and physical activity units, respectively, 23 and 7 h wk⁻¹.

During the study, neither group reported participating in any organized or recreational physical activity outside the protocol; only the IT group performed the interval training.

The physical activity classes in which the participants were enrolled were designed with a primary focus on the methods of teaching physical education, not for promoting fitness. All participants, according to self-reports, strictly adhered to the study's physical activity restrictions, thereby increasing the likelihood that the changes observed resulted from the interval training instead of other outside influences.

Procedures

Anthropometric Measurements. Body mass (BM) and body composition were estimated using a bioelectrical impedance floor scale (TBF-300 Body Fat Monitor/Scale Analyzer, Tanita, Japan) calibrated in accordance with manufacturer guidelines before each test session. One hour after a light breakfast, participants voided their bladder and bowels and,

TABLE 3. Comparison of aerobic parameters before and after HIIT intervention.*†

Variable	Pretraining		Posttraining	
	IT-group	C-group	IT-group	C-group
$\dot{V}O_{2\max}$ (L·min ⁻¹)	4.1 ± 0.3	3.9 ± 0.3	4.5 ± 0.4‡§	3.9 ± 0.4
$\dot{V}O_{2\max}$ (ml·kg ⁻¹ ·min ⁻¹)	50.1 ± 3.1	48.2 ± 4.7	55.6 ± 5.5‡§	48.5 ± 3.9
Power at $\dot{V}O_{2\max}$ (W)	360 ± 35	330 ± 27	406 ± 38‡§	347 ± 27
Power at $\dot{V}O_{2\max}$ (W·kg ⁻¹)	4.4 ± 0.3	4.0 ± 0.5	5.0 ± 0.3	4.3 ± 0.4
Power at AT (W)	277 ± 44	255 ± 34	307 ± 29	250 ± 31
Power at AT (W·kg ⁻¹)	3.4 ± 0.5	3.1 ± 0.5	3.7 ± 0.2‡	3.0 ± 0.4
$\dot{V}O_{2\max}$ at AT (ml·kg ⁻¹ ·min ⁻¹)	39.6 ± 4.7	38.6 ± 5.3	43.4 ± 2.7‡§	36.5 ± 2.4§
AT as percentage of $\dot{V}O_{2\max}$	79.0 ± 7.5	79.6 ± 5.5	78.5 ± 6.7	75.7 ± 7.1

* $\dot{V}O_{2\max}$ = maximal oxygen uptake; AT = anaerobic threshold; IT = interval training group; C = control group; HIIT = high-intensity interval training.

†Values are mean ± SD.

‡Significant differences ($p \leq 0.05$) vs. C-group.

§Significant differences ($p \leq 0.05$) vs. pretraining.

TABLE 4. Comparison of anaerobic parameters before and after HIIT intervention.*†

Variable	Pretraining		Posttraining	
	IT-group	C-group	IT-group	C-group
Work output ($\text{J} \cdot \text{kg}^{-1}$)	261.1 $\pm$ 11	262.9 $\pm$ 15	272.3 $\pm$ 13‡§	266.7 $\pm$ 13
Phosphagenic work ($\text{J} \cdot \text{kg}^{-1}$)	76.8 $\pm$ 11	80.6 $\pm$ 13	75.5 $\pm$ 8.9	80.8 $\pm$ 14.1
Phosphagenic work (% of total work)	29.5 $\pm$ 4.7	30.7 $\pm$ 4.7	27.8 $\pm$ 2.1	30.4 $\pm$ 5.8
Nonphosphagenic component of work ($\text{J} \cdot \text{kg}^{-1}$)	184.3 $\pm$ 16.0	182.2 $\pm$ 16.0	196.8 $\pm$ 16.7	186.0 $\pm$ 21.9
Nonphosphagenic component of work (% of total work)	70.5 $\pm$ 4.7	69.3 $\pm$ 4.2	72.1 $\pm$ 3.7	69.5 $\pm$ 5.8
Mean power ($\text{W} \cdot \text{kg}^{-1}$)	8.7 $\pm$ 0.3	8.7 $\pm$ 0.5	9.0 $\pm$ 0.4‡§	8.7 $\pm$ 0.4
Peak power ($\text{W} \cdot \text{kg}^{-1}$)	11.4 $\pm$ 0.4	10.9 $\pm$ 1.0	11.8 $\pm$ 0.7	10.9 $\pm$ 0.8
Max power ($\text{W} \cdot \text{kg}^{-1}$)	11.3 $\pm$ 0.4	10.9 $\pm$ 0.9	11.7 $\pm$ 0.7	10.8 $\pm$ 0.9
Fatigue index (%)	23.7 $\pm$ 3.6	20.0 $\pm$ 5.7	19.2 $\pm$ 2.4	18.6 $\pm$ 4.5
Time to max power (s)	5.1 $\pm$ 1.2	5.4 $\pm$ 1.0	4.2 $\pm$ 0.6‡	4.9 $\pm$ 0.9
Time at max power (s)	4.6 $\pm$ 1.2	4.6 $\pm$ 1.0	3.3 $\pm$ 0.7	3.7 $\pm$ 1.1

*IT = interval training group, C = control group; nonphosphagenic component of work – glycolytic work and oxidative phosphorylation.

†Values are means $\pm$ SD.

‡Significant differences ($p \leq 0.05$) vs. C-group.

§Significant differences ($p \leq 0.05$) vs. pretraining.

clad only in briefs, underwent duplicate measures while in the standing position recommended by the manufacturer guidelines. The average of the 2 values was used for final analysis.

Aerobic Power Measurement. To determine $\dot{V}\text{O}_2\text{max}$ and anaerobic threshold (AT), participants performed a graded cycle ergometry test on an electromagnetically braked cycle ergometer (ER 900 Jaeger, Germany/Viasys Health Care). The ergometer seat height was individually adjusted to attain a 5° bend in the knee at the lowest point in the pedal revolution. Participants were allowed a 5-minute warm-up period at an intensity of 1.5 $\text{W} \cdot \text{kg}^{-1}$ with a pedaling cadence of 60 rpm. Immediately after the warm-up, the participants began $\dot{V}\text{O}_2\text{max}$ testing by cycling at increasingly difficult workloads in which resistance was increased by 25 $\text{W} \cdot \text{min}^{-1}$ until the participant reached the point of volitional

exhaustion. The recovery was passive with the participant in a seated position. Breath-by-breath pulmonary gas exchange was measured (Oxycon-Pro, Jaeger-Viasys Health Care, Hochberg, Germany) throughout the $\dot{V}\text{O}_2\text{max}$ test; the O_2 and CO_2 analyzers were calibrated before each test using standard gases of known concentrations in accordance with manufacturer guidelines. The nonlinear increase in ventilation (ventilatory threshold) was used to determine AT. Heart rates were monitored continuously by telemetry (S-625, Polar Electro-Oy, Finland) during each test session and the first 5 minutes of passive recovery in a seated position.

Anaerobic Power Measurement. Participants performed the WAnT on a mechanically braked cycle ergometer (884E Sprint Bike, Monark, Sweden) according to the procedures described previously (2,23). The testing session started with

TABLE 5. Blood lactate levels after Wingate tests before and after HIIT training.*†

Variable	Pretraining		Posttraining	
	IT-group	C-group	IT-group	C-group
LA at rest ($\text{mmol} \cdot \text{L}^{-1}$)	1.0 $\pm$ 0.3	1.8 $\pm$ 0.3	1.0 $\pm$ 0.3	1.5 $\pm$ 0.3
LA after (5 min) WAnT ($\text{mmol} \cdot \text{L}^{-1}$)	15.5 $\pm$ 0.8	15.0 $\pm$ 0.6	12.4 $\pm$ 0.5‡§	15.5 $\pm$ 0.7
LA after (15 min) WAnT ($\text{mmol} \cdot \text{L}^{-1}$)	11.4 $\pm$ 1.7	13.6 $\pm$ 1.6	9.4 $\pm$ 2.1‡	12.1 $\pm$ 2.2

*LA = lactic acid, WAnT = Wingate anaerobic test, IT = interval training group, C = control group.

†Values are means $\pm$ SD.

‡Significant differences ($p \leq 0.05$) vs. C-group.

§Significant differences ($p \leq 0.05$) vs. pretraining.

a standardized 5-minute warm-up cycling at $1.0 \text{ W} \cdot \text{kg}^{-1} \cdot \text{BM}^{-1}$ including 2 all-out sprints lasting 3–5 seconds each against the resistance used for the actual test. After a 5-minute rest, the WAnT began from a stationary starting position with the participant seated and the right pedal at approximately 45° as previously described (23,30). The participants were instructed to accelerate to their maximal pedaling rate and were verbally encouraged to maintain this pedaling cadence as long as possible throughout the 30-second test. Only during the first few, initial movements of the test, was each participant able to pedal in a standing position; this helped overcome the resistance and achieve the maximal pedaling rate. The remaining part of the test was performed with the participant in a seated position. A flywheel resistance equaling $0.075 \text{ kg} \cdot \text{kg} \cdot \text{BM}^{-1}$ (corresponding to 7.5% of each individual's BM) was applied at the onset of the WAnT (10). After test termination, the participants were supervised during a 15-minute recovery in a seated or supine position. All tests were performed at similar times in the morning at least 2 hours after a light breakfast.

The computer software automatically calculated mechanical power by multiplying braking force by velocity (pedaling cadence) for each 5-second segment of the WAnT. Four other indices describing the participant's WAnT performance were also calculated programmatically. Peak power (PP), defined as the highest mechanical power achieved at any stage of the test, typically occurred in the first few seconds of the WAnT and was used to represent the explosive characteristics of the participant's muscle power. Mean power (MP), the average power sustained throughout the 30-second period, was used to represent the average local muscle endurance throughout the WAnT. The fatigue index (FI) or degree of power drop from PP to the lowest power at the end of the test, and was expressed in $\text{W} \cdot \text{s}^{-1}$, $\text{W} \cdot \text{kg} \cdot \text{BM}^{-1}$ and as a percentage of PP. Total work (W_{tot}) was used to describe muscle endurance and was calculated by multiplying MP by 30 seconds (duration of the WAnT protocol) and expressed in Joules (J) or $\text{J} \cdot \text{kg} \cdot \text{BM}^{-1}$ (2). It was assumed that PP primarily reflected the participant's ability to convert alactic (phosphagenic) energy, whereas MP mainly reflected the combined rate of lactic (glycolytic-nonphosphagenic), phosphagenic, and aerobic energy conversion.

In the present study, the phosphagenic and nonphosphagenic components of work were calculated as previously described (38). This mathematical

method is based on calculating the total area under the curve from the beginning to the end of WAnT (from 0 to 30 seconds on the horizontal axis). The total diagram area (Figure 1) is subsequently separated into 2 components, divided at the perpendicular line from point A on the power course curve to point B on the time axis; point B corresponds to the onset of power drop from PP (point A). The first of the 2 parts corresponds to the phosphagenic ($W_{\text{anaphosph}}$) component of work, and the second one represents the nonphosphagenic ($W_{\text{ananonphosph}}$) component of work. This method provides an estimation of the proportion of $W_{\text{anaphosph}}$ and $W_{\text{ananonphosph}}$ contribution to the total anaerobic energy turnover during the WAnT. $W_{\text{anaphosph}}$ and $W_{\text{ananonphosph}}$ may be expressed as percentages of total anaerobic work (W_{anatot}).

Blood Lactate Analysis. Blood samples were collected from an antecubital vein as part of the baseline and follow-up WAnT procedures with samples taken before warm-up and 5 and 15 minutes after WAnT. Immediately after collection, the blood was deproteinized by the addition of ice cold 0.4 M perchloric acid. After being thoroughly mixed, the samples were centrifuged at $12,000g$ for 10 minutes. Blood LA was determined using a standard Randox (Crumlin, United Kingdom) kit based on the LA oxidase method (LC2389); assays were performed on a Cecil CE9200 spectrophotometer (Cambridge, United Kingdom).

Training Intervention. Participants in C continued with their regularly scheduled classes and activities; Participants in IT, also in the same academic program, continued with these same regularly scheduled activities in addition to the training protocol. The IT group met every Monday, Wednesday, and Friday for 6 weeks. Each training session began at 0900, 1

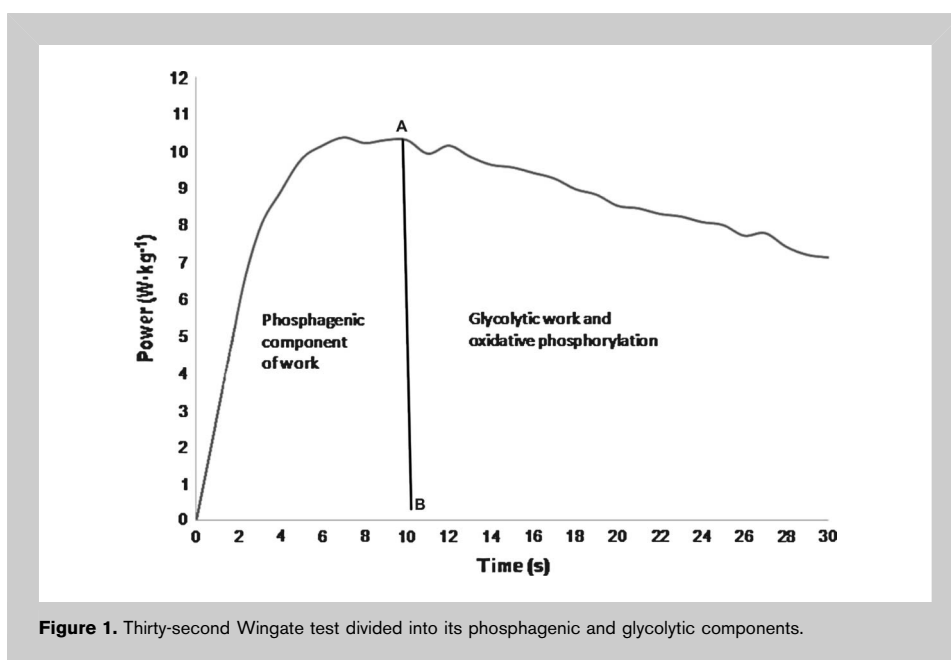


Figure 1. Thirty-second Wingate test divided into its phosphagenic and glycolytic components.

hour after the participants ate a light breakfast. The same research assistant who controlled the flywheel resistance and also timed the interval and recovery periods supervised all sessions. The ergometer (884E Sprint Bike, Monark, Sweden) seat height was individually identified for each participant (5° bend in the knee at the lowest point in the revolution) and maintained throughout the intervention period. Each training session began with a 5-minute warm-up at an intensity approximating 30% of maximal aerobic power ($30\% \dot{V}O_{2\max}$). Six 90-second cycling bouts at $80\% \dot{V}O_{2\max}$ were then performed at a cadence of 60 rpm. Each 90-second bout was followed by 180 seconds of passive rest for a work-to-rest ratio of 1:2. All laboratory sessions were performed in ambient conditions of $20\text{--}22^\circ\text{C}$ and 60% humidity.

Statistical Analyses

Statistical analyses were performed using Statistica 8.0 for Windows. A 2 (group) $\times$ 2 (time) repeated-measures analysis of variance was used to investigate the significance of differences between groups and pre and posttraining sessions. The normality of data was tested using the Shapiro–Wilks W test. To assess the influence of initial values on power output, Pearson correlations were performed to compare the relationship between the pre and postintervention values of power at $\dot{V}O_{2\max}$ in absolute and relative terms. Statistical significance was set at $p \leq 0.05$ for all analyses. Targeting statistical power at $\beta = 0.8$ with an effect size of 0.7, a sample size estimate of 12 participants per group was determined (39).

RESULTS

All participants completed the study, and no adverse events were reported. The Shapiro–Wilks W test was significant ($p < 0.05$). At baseline, there were no significant differences in basic anthropometric characteristics between groups (Table 2) or in aerobic (Table 3) and anaerobic performance (Table 4). There were significant ($p < 0.05$) main effects of group and time for several aerobic (Table 3) and anaerobic parameters (Table 4). Additionally, as shown in Table 5, there was a significant main effect of group for LA at the 5 and 15 minutes post-WAnT ($\text{mmol}\cdot\text{L}^{-1}$) marks for IT; $p < 0.05$. Significant interaction effects were noted for absolute ($\text{L}\cdot\text{min}^{-1}$) and relative ($\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) $\dot{V}O_{2\max}$, absolute (W) and relative

($\text{W}\cdot\text{kg}^{-1}$) power at $\dot{V}O_{2\max}$, power at AT, $\dot{V}O_2$ at AT ($\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$), work output ($\text{J}\cdot\text{kg}^{-1}$), and MP ($\text{W}\cdot\text{kg}^{-1}$) with those in IT, but not C, experiencing improvements; $p < 0.05$ (Tables 3 and 4).

The 6-week HIIT program favorably influenced the aerobic and anaerobic performances for the IT group. Absolute and relative $\dot{V}O_{2\max}$ increased ($p < 0.05$) by an average of $0.5 \text{ L}\cdot\text{min}^{-1}$ (13%) and $5.5 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ (11%), respectively (Table 3). The pre and posttraining correlations between aerobic performance expressed in absolute ($r = 0.93$) and relative ($r = 0.86$; Figure 2) power terms for IT indicate that their training effect was similar regardless of their respective initial values. Improvements ($p < 0.05$) of 29.5 W and $0.29 \text{ W}\cdot\text{kg}^{-1}$ in absolute and relative power at AT, respectively, and $3.8 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ in $\dot{V}O_{2\max}$ at AT were recorded for IT on average (Table 3). Mean values for absolute and relative power at AT decreased slightly for C between the pre and posttraining periods.

As indicated by data presented in Table 4, the IT group improved in several anaerobic performance variables when compared to C. Work output ($\text{J}\cdot\text{kg}^{-1}$) increased 4.3% for IT but only by 1.4% for C. Although the average nonphosphagenic component of work ($\text{J}\cdot\text{kg}^{-1}$) increased for both groups, the change was larger for IT ($12.5 \text{ J}\cdot\text{kg}^{-1}$; 6.8%); this was also reflected through an increase in the mean difference of the relative contribution of nonphosphagenic component of work to the total amount of work performed (1.6%) and a corresponding decrease of 1.7% in the relative contribution

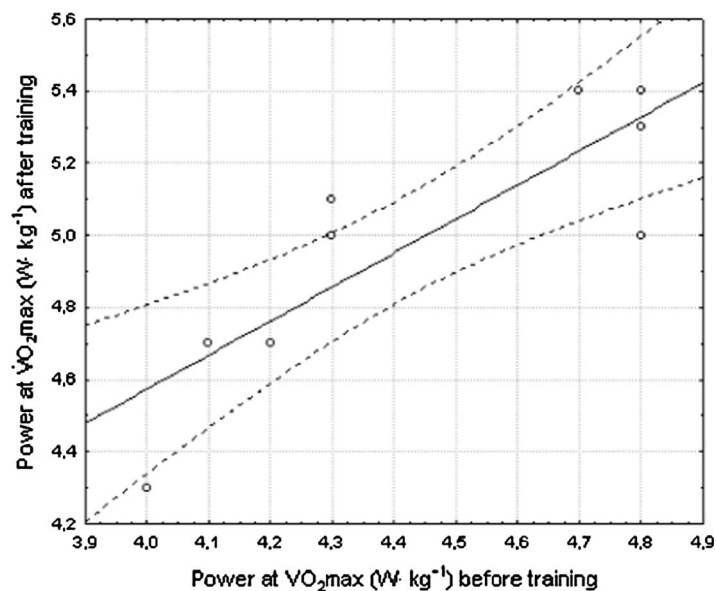


Figure 2. Relationship between baseline and follow-up values for relative power ($\text{W}\cdot\text{kg}^{-1}$) at $\dot{V}O_{2\max}$ for those in interval training group. Line of best fit and 95% confidence intervals are presented as the solid and dashed lines, respectively.

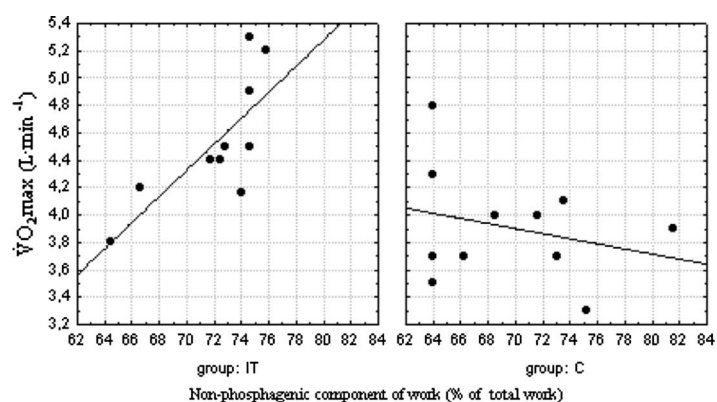


Figure 3. Relationship between $\dot{V}O_{2\max}$ (L·min⁻¹) and percent contribution of nonphosphagenic work after high-intensity interval training protocol for training and control groups.

of the phosphagenic system for IT. Only in IT group, was a significant correlation between $\dot{V}O_{2\max}$ and nonphosphagenic component of total work (%) observed (Figure 3). The relative contribution of the anaerobic energy sources changed by less than 0.5% for C. Mean power improved, on average, for IT only. Both groups attained PP more quickly at the end of the 6-week period; however, the decrease in time to attain PP was 0.9 seconds for IT compared to 0.5 seconds for C. On the other hand, the mean difference in the ability to sustain PP revealed that C was able to maintain PP production 0.4 seconds longer than was IT during follow-up testing. Both groups demonstrated that LA values increased from resting values to 5 minutes postexertion with a decline after an additional 10 minutes of rest (Table 5). Significant ($p < 0.05$) improvements in LA response were noted only for IT. Compared to baseline, LA values for IT when 5 and 15 minutes into recovery were 3.1 and 2.0 mmol·L⁻¹ lower, respectively, after HIIT training.

DISCUSSION

The major finding of this study was that 27 min·wk⁻¹ of HIIT performed on a stationary cycle ergometer at a 1:2 work-to-rest ratio throughout a 6-week period significantly improved numerous variables associated with aerobic and anaerobic performance but decreased LA formation. Our protocol requiring the participants to cycle at 80% of their baseline $\dot{V}O_{2\max}$ with a 1:2 work-to-rest ratio was tolerated by all in the IT group.

Unlike in the protocol used by Laursen et al. (27) in which they adjusted the training workload after a midpoint reassessment, our workload remained constant over the 6-week intervention. We observed an 11% increase in $\dot{V}O_{2\max}$ (ml·kg⁻¹·min⁻¹) for the IT group, whereas Laursen et al. (27) reported an increase of 6%. The 5% difference in final outcome between the 2 studies may be the result of the initial cardiorespiratory training levels of the participants and the

duration of the interventions. There are several other studies showing increases in the maximal aerobic capacity ($\dot{V}O_{2\max}$) after interval training (16,29).

Our results also support the findings of Creer et al. (16) and MacDougall et al. (29) even though these 2 studies used a 1:8 work-to-rest ratio for a sample of trained cyclists and healthy college students, respectively. Although we noted an average improvement in $\dot{V}O_{2\max}$ of 5.5 ml·kg⁻¹·min⁻¹ after a 6-week protocol using a 1:2 work-to-rest ratio, MacDougall et al. (29) reported an improvement of 3.5

ml·kg⁻¹·min⁻¹ after a 7-week protocol. Creer et al. (16) assigned their trained male cyclists into endurance-only and short-sprint-plus-endurance intervention groups; they noted an increase in $\dot{V}O_{2\max}$ of 0.2 L·min⁻¹ for both groups. As shown in Table 3, we reported an increase of 0.41 L·min⁻¹ for the IT group. Again, the initial conditioning level of the participants compared to ours cannot be overlooked when comparing the results from these 2 studies. In terms of indicators of aerobic endurance, those in the IT group of the present study increased their $\dot{V}O_2$ consumption at AT by 3.5 ml·kg⁻¹·min⁻¹; however, AT as a percentage of $\dot{V}O_{2\max}$ at posttesting did not differ from the baseline values. Similarly, Creer et al. (16) reported no change in the percentage of maximal heart rate at which ventilatory threshold was observed.

To investigate the effects of HIIT on anaerobic capacity, a Wingate test was performed. We noted that total work output was significantly higher in the IT group. As performance in WAnT is dependent on phosphagenic, glycolytic and, partially, oxidative metabolism, these data indicate that at least one of these energetic systems had improved after interval training. Our data indicate that the phosphagenic work was the same in both groups but the nonphosphagenic component of work tended to be higher in the IT group. Recent evidence suggests that during exercise lasting up to 30 seconds, a substantial amount of energy is derived from aerobic metabolism. Therefore, we consider that improvement in WAnT could be a result of increase in glycolysis and oxidative phosphorylation. To evaluate the influence of HIIT on glycolysis, we measured the blood LA concentration after WAnT. Blood LA values taken after the WAnT were lower on average for our IT group (Table 5). This 3-mmol·L⁻¹ decrease in concentration at the 5-minute mark indicates that LA formation was significantly lower in the IT group. On the other hand, when we compared changes in LA concentration 15 minutes after WAnT, there was no difference between groups. These data suggest that 6 weeks of HIIT did not influence the rate of LA removal.

Our data are in agreement with those of Burgomaster et al. (11); their sprint cycling group had lower rates of glycogenolysis and, hence, muscle LA accumulation because the men in that group also had an increased ability to oxidize pyruvate. Moreover, 8 weeks of endurance training was sufficient to increase nicotinamide adenine dinucleotide reduced form (NADH) shuttle enzymes levels by approximately 50%; this enzymatic system competes with LA dehydrogenase for NADH (35). Because we had no capabilities of measuring NADH shuttle enzyme activity, we can only speculate that our training protocol decreased LA formation by increasing oxidative capabilities of cytoplasmic NADH via the shuttle enzymes. Indeed, recent evidence suggests that 13% of energy during a 10-second sprint and 27% of energy during a 20-second sprint is generated aerobically (9,24). Postintervention results for the IT group in the present study revealed an increased nonphosphagenic component of total work performed during WAnT (Figure 3). All together these data indicate that an increase in total work output during WAnT is a result of an increase in aerobic metabolism. This conclusion is in direct opposition to the studies reporting an increase in anaerobic metabolism as a result of HIIT. For example, Creer et al. (16) noted higher LA levels for the short-sprint-plus-endurance training group during posttraining compared to baseline for each of the 4 sprints and at 3- and 6-minutes into recovery after the last sprint. Using a sample of college students similar to those in the present study, MacDougall et al. (29) noted increases in oxidative and glycolytic enzymes after a 7-week intervention of 30-second sprint sessions 3 times per week. Our data indicate that 6 weeks of an HIIT protocol with a 1:2 work-to-rest ratio is an effective means to improve aerobic capacity and that the increased performance during WAnT is a result of increased aerobic capacity.

PRACTICAL APPLICATIONS

For coaches, athletes, and fitness practitioners who need to quickly improve maximal aerobic and anaerobic cycling performance, using our 1:2 work-to-rest HIIT protocol should produce notable and favorable changes with minimal time invested in cycling-specific training per week. This protocol may shorten preseason training requirements or provide a time-efficient method to maintain in- and postseason performance capacities of men similar to those in our study. For the tennis players who lost 4–15% of their aerobic capacity during a 3–6 week break (31) and for other athletes, our HIIT protocol with a 1:2 work-to-rest ratio may be a time-efficient means to counter detraining during the off season.

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the authors, because there are no professional relationships with companies or manufacturers who will benefit from the results of the present study. The results presented herein do not constitute an endorsement of the protocol by the NSCA nor by the authors as the only method by which to quickly show such improvements in aerobic and anaerobic performance. Disclosure of funding: There was no external funding for this project.

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